A Multifaceted Look at Starlink Performance

Nitinder Mohan∗Andrew E. Ferguson∗Hendrik Cech∗
Technical University of Munich
GermanyThe University of Edinburgh
United KingdomTechnical University of Munich
Germany

In recent years, Low-Earth Orbit (LEO) mega-constellations have ushered in a new era for ubiquitous Internet access we analyze Starlink global performance relative to terrestrial cellular networks Second, The researchers do examine Starlink's ability to support real-time latency and bandwidth-critical applications by analyzing the performance of (i) Zoom conferencing, and (ii) Luna cloud gaming, comparing it to 5G and fiber.
They perform measurements from Starlink-enabled RIPE Atlas probes to shed light on the lastmile access and other factors affecting its performance. Also conduct controlled experiments from Starlink dishes in two countries and analyze the impact of globally synchronized "15-second reconfiguration intervals" of the satellite links that cause substantial latency and throughput variations LEO networks consist of mega-constellations with thousands of satellites orbiting at 300–2000 km altitudes, promising ubiquitous low latency coverage worldwide Starlink from SpaceX stands out with its expansive fleet of 4000 satellites catering to 2M+ subscribers across 63 countries his paper addresses this knowledge gap and provides the first comprehensive multi-faceted measurement study on Starlink The analysis showed that Starlink is comparable to cellular for supporting real-time applications (Like Zoom and Luna cloud gaming), though this varies based on proximity to ground infrastructure. The study shows Starlink inter-satellite connections helping remote users achieve better Internet service than terrestrial networks.

Documen2t